Manoa Randrianarisoa

TECHNICAL ARCHITECT & SENIOR LEAD FULL-STACK

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SKILLS

FRONTEND
React 18/19, TypeScript, Next.js, React Native, Tailwind CSS, Vite, Zustand

BACKEND
Node.js, NestJS, Express 5, PostgreSQL, Supabase, Firebase, SQLite (WAL), GraphQL, REST

ARCHITECTURE
Multi-tenant SaaS, RBAC, offline-first, Electron, PWA / IndexedDB, Docker, CI/CD, Linux

INTEGRATION & AUTOMATION
Make, n8n, webhooks, Sage ERP APIs, BI dashboards

AI & DATA
Python, OpenCV, TensorFlow, algorithms

GOVERNANCE
Agile / Scrum, functional framing, SQL acceptance, traceability matrix, mentoring

EDUCATION

MSc Electronics, Computer Science & AI
ISPM · 2022 — 2024
Defended and validated; diploma pending issuance. Computer vision, Deep Learning, signal processing.

BSc Electronics & Computer Science
ISPM · 2018 — 2022
Software engineering, system architectures, concurrent programming, networks.

Baccalauréat série D (science track)
La Pépite d'Or private high school, Antananarivo · 2018

LANGUAGES

French (fluent) · English (professional, technical interviews) · Malagasy (native)

CONTRACTUAL FRAMEWORK

Missi Innovation — sole proprietorship registered in Madagascar. Invoicing in MGA or EUR with full statutory details; tax identifiers (NIF / STAT) provided on request, within the contractual framework. Direct freelance, international umbrella company or agency subcontracting.

AVAILABILITY

Immediate — remote worldwide or on-site. Time zone UTC+3 (2 hours ahead of Paris in winter). Full-time, part-time or one-off framing.

PROFILE

Technical architect and lead full-stack, **8 years of practice**: system resilience (**offline-first**), **multi-tenant SaaS**, mentoring remote teams. From functional framing to production, I design measurable systems and train the teams that keep them alive.

PROFESSIONAL EXPERIENCE

Independent consultant / Technical architect
2025 — present
Missi Innovation · Antananarivo — international remote

- **ANKA (AFD)**: offline-first ESG field collection — 45 traced requirements, React + Dexie PWA, NestJS API, 14 days offline.
- **ResHPro**: multi-tenant hospitality SaaS (POS, inventory, RBAC) — React 18, PostgreSQL, RLS isolation (ADR-002).
- **Study-Agent**: school ERP in production (Manjary Soa High School) — Electron, SQLite WAL, Supabase, offline-safe entry; Make & n8n.
- **Maboo**: frontend lead, headless storefront React 19 / React Router 7 SSR (client code not public).

Technical architect & mobile lead
2025 — 2026
Given Day Tech · France (non-profit) — remote

- Mentoring **4 developers** remotely: React Native + Node.js application architecture, code reviews, mentoring.

Senior full-stack developer
2025 — present
ToutPourSaGloire.com · international — remote

- Meilisearch search engine and Next.js overhaul: **40% less loading time** (measured in development environment).

CMS developer & UI/UX designer
2023 — 2024
RICSM · United States — remote, contracted via IKS Antananarivo

- Squarespace sites for international clients, Figma mockups, performance & SEO; successful launches (ricsm.org).

Full-stack & OCR pipeline engineering intern
May — Oct 2022
EQIMA Solution · Ivandry, Antananarivo

- OCR pipeline (OpenCV) into the Sage ERP API: **70% less processing time** for accounting teams.

Other experience
2019 — 2023
Teaching & customer service — Antananarivo

- Part-time teacher (2020 — 2023): computer science, electronics, applied mathematics — private lessons and groups.
- ADSL technical service, SFR France customers (Oct — Dec 2019) — Intelcia Tanjombato, level-1 support.

STUDY PROJECTS — ISPM

2024	**Aizah — augmented-reality navigation.** Indoor navigation through AR and AI, precise localization. Vue · React Native · Python · OpenCV · TensorFlow
2023	**Biometric attendance & staff tracking.** Fingerprint authentication, real time, automatic reporting. React · Node.js · Prisma · PostgreSQL · Arduino
2021-22	**Geolocation of mobile assets.** Real-time GPS tracking with optimized accuracy. Ionic · Laravel · REST API
2020	**Automated drink vending machine.** Remote control through a mobile application. Angular · Python · Arduino
2019	**Laser data transmission.** Laser modulation with a verification protocol. JavaScript · Python · C · Arduino

Year-long projects carried in groups of 3 to 5: jack-of-all-trades role, often full-stack, group lead on some.

ARCHITECTURE DECISIONS

ADR-001	**Offline-first SQLite (WAL) + Supabase.** Adopted, in production: withstand unstable network and power.
ADR-002	**Multi-tenant isolation via PostgreSQL RLS.** Framed, migration in progress: preferred over multi-database splitting.
ADR-003	**React Router 7 SSR with streaming.** Deployed (Maboo): non-critical blocks streamed after critical content.